

Excel Power Query

Power Query

Power Query (known as **Get & Transform** in Excel) allows you to import or connect to external data and then shape that data—for example, by removing a column, changing a data type, or merging tables—in ways that meet your needs. After shaping the data, you can load your query into Excel to create charts and reports. Periodically, you can refresh the data to keep it up to date. **Power Query** is available on three Excel applications: **Excel for Windows**, **Excel for Mac**, and **Excel for the Web**.

The Four Phases of Power Query

1. Connect

Make connections to data in the cloud, on a service, or locally.

2. Transform

Shape data to meet your needs while the original source remains unchanged.

3. Combine

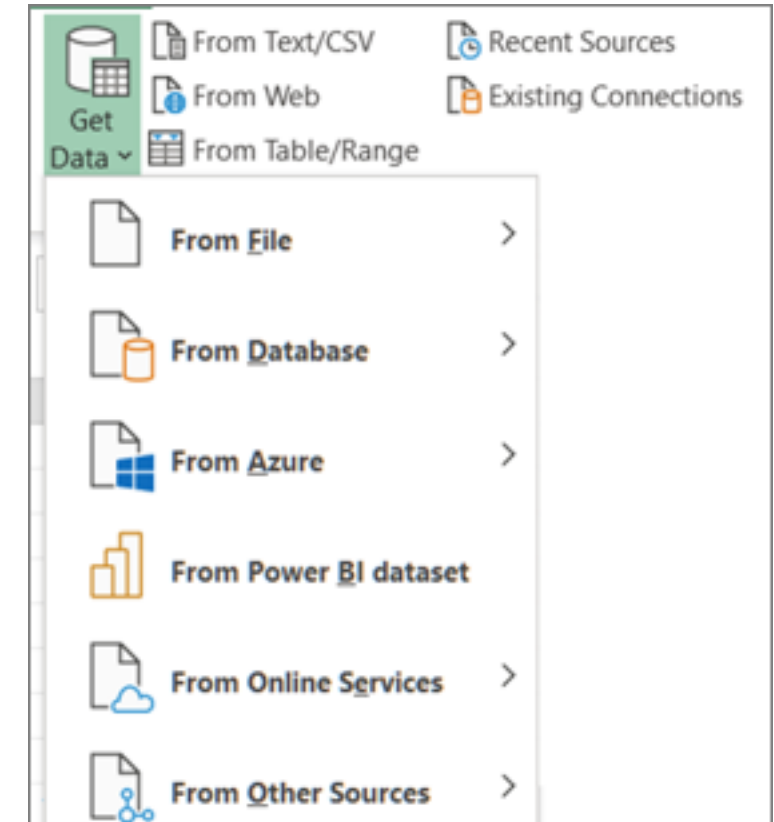
Integrate data from multiple sources to get a unique view of the data.

4. Load

Complete your query and load it into a worksheet or Data Model, and periodically refresh it.

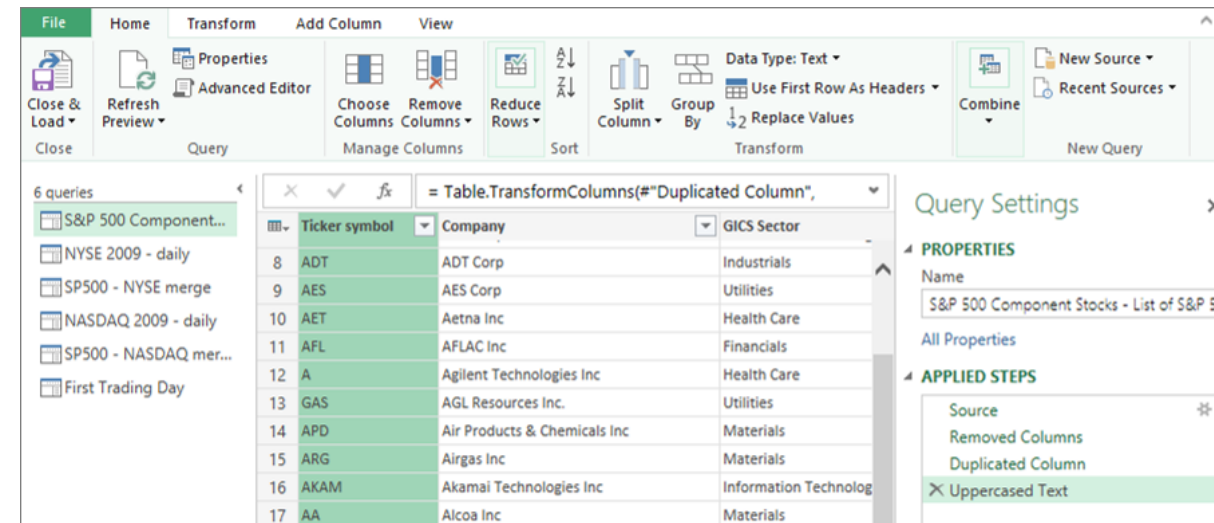


- You can use **Power Query** to import data from a single data source, such as an Excel workbook, or from multiple databases, feeds, or services scattered across the cloud. Data sources include data from the Web, files, databases, Azure, or even Excel tables in the current workbook. With **Power Query**, you can then bring all those data sources together using your own unique transformations and combinations to uncover insights you otherwise wouldn't have seen.
- Once imported, you can refresh the data to bring in additions, changes, and deletions from the external data source.



Transform

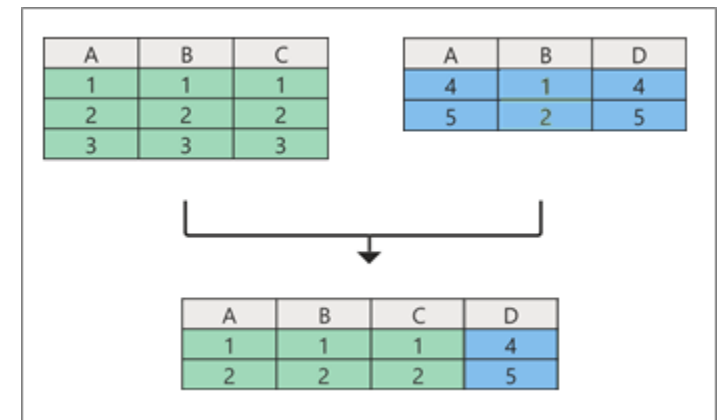
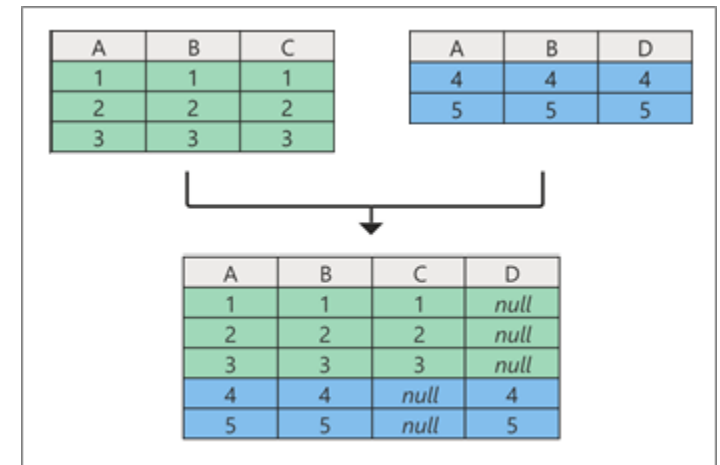
- **Transforming data** means modifying it in some way to meet your data analysis requirements. For example, you can remove a column, change a data type, or filter rows. Each of these operations is a data transformation. This process of applying transformations (and combining) to one or more sets of data is also called **shaping data**.
- Think of it this way: a vase starts as a lump of clay that one shapes into something practical and beautiful. **Data** is the same; it needs shaping into a table that is suitable for your needs and enables attractive reports and dashboards.
- **Power Query** uses a dedicated window called the **Power Query Editor** to facilitate and display data transformations.
- You can open the **Power Query Editor** by selecting **Launch Query Editor** from the **Get Data** command in the **Get & Transform Data** group, but it also opens when you connect to a data source, create a new query, or load a query.



Append: An append operation creates a new query that contains all rows from a first query followed by all rows from a second query. You can perform two types of append operations:

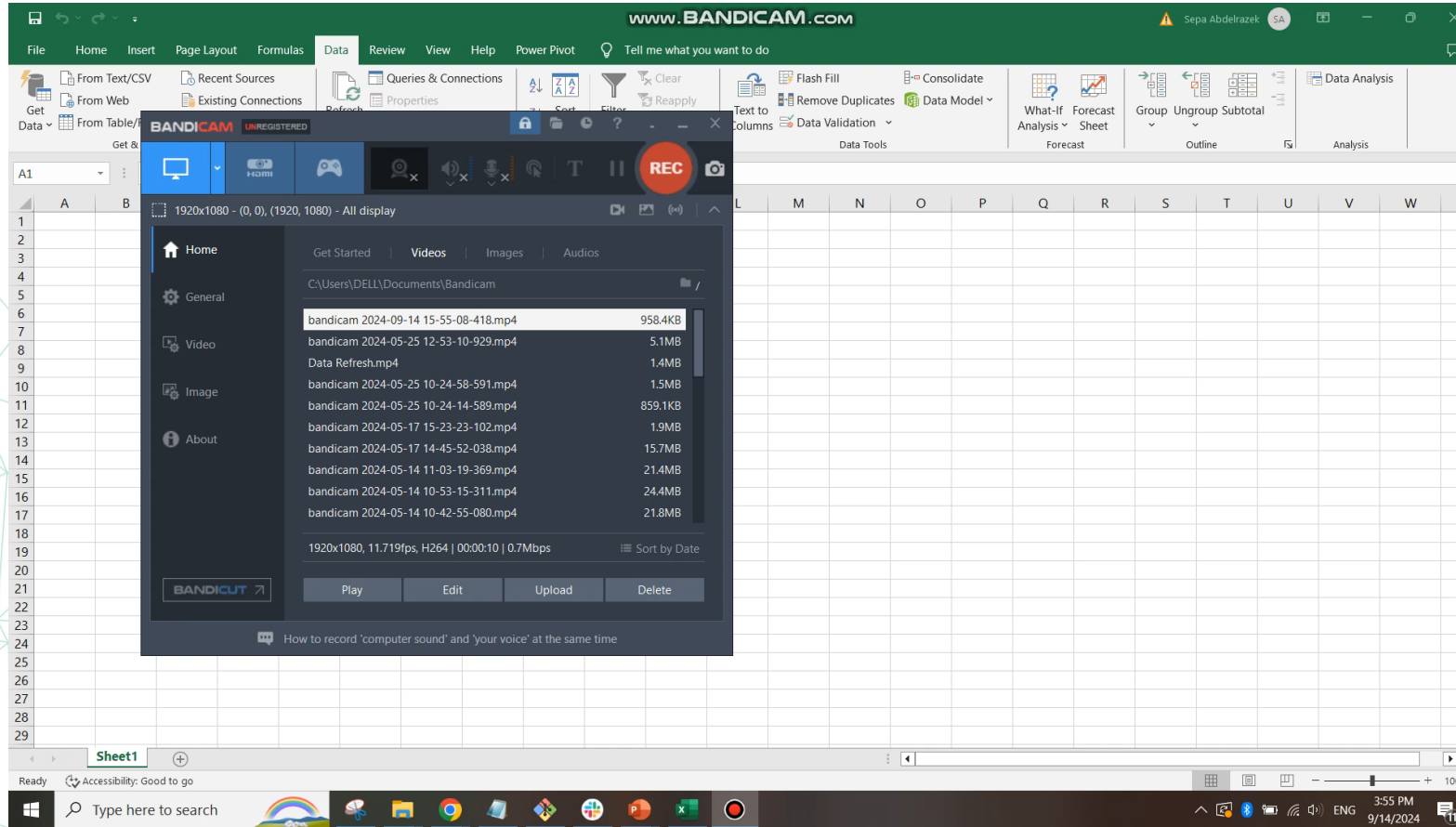
- **Intermediate Append:** Creates a new query for each append operation.
- **Inline Append:** Appends data to your existing query until you reach a final result.

Merge: A merge operation creates a new query from two existing queries. This one query contains all columns from a primary table, with one column serving as a navigation link to a related table. The related table contains all rows that match each row from a common column value in the primary table. Furthermore, you can expand or add columns from a related table into the primary table.



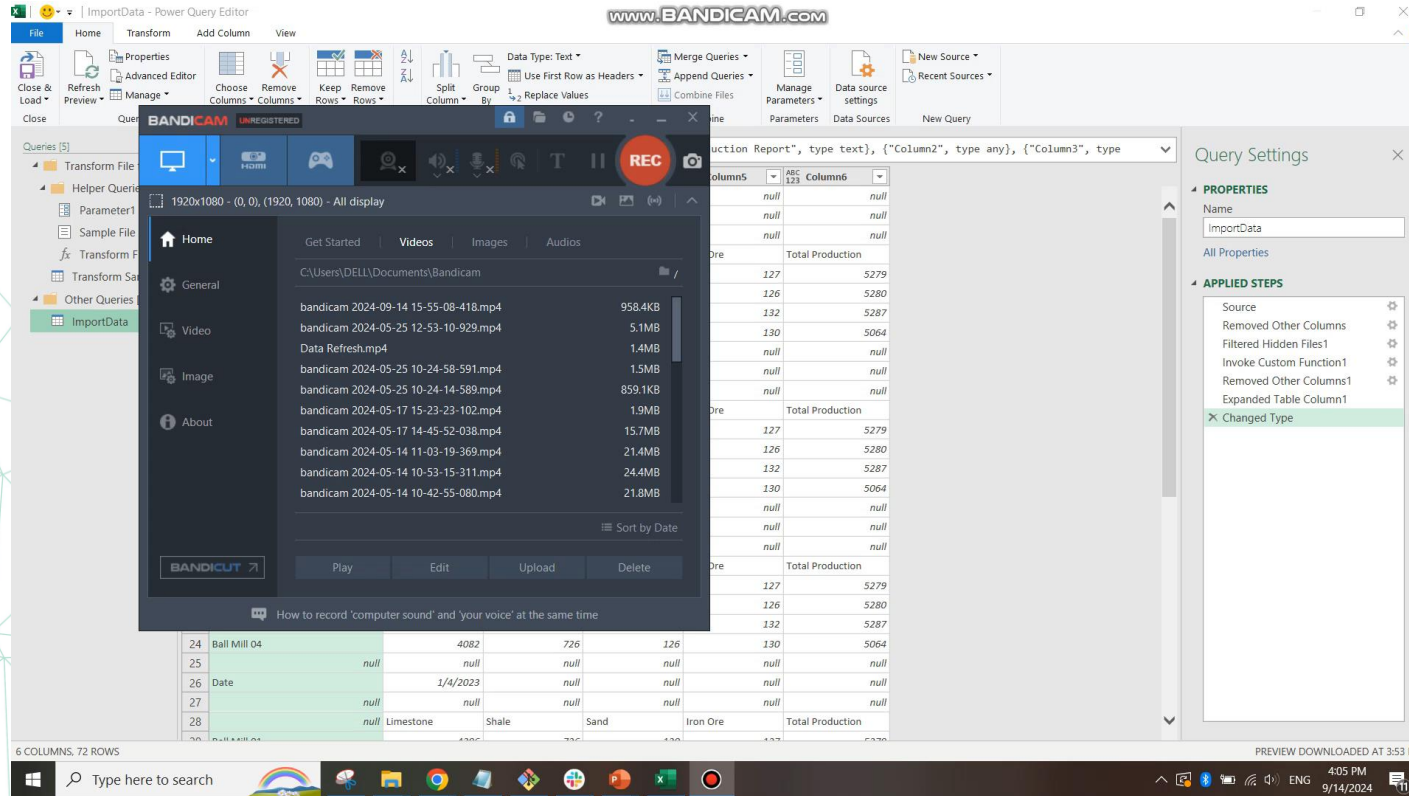
Step 1 Data Loading

Follow from the Video



Step 2 Removing unnecessary data

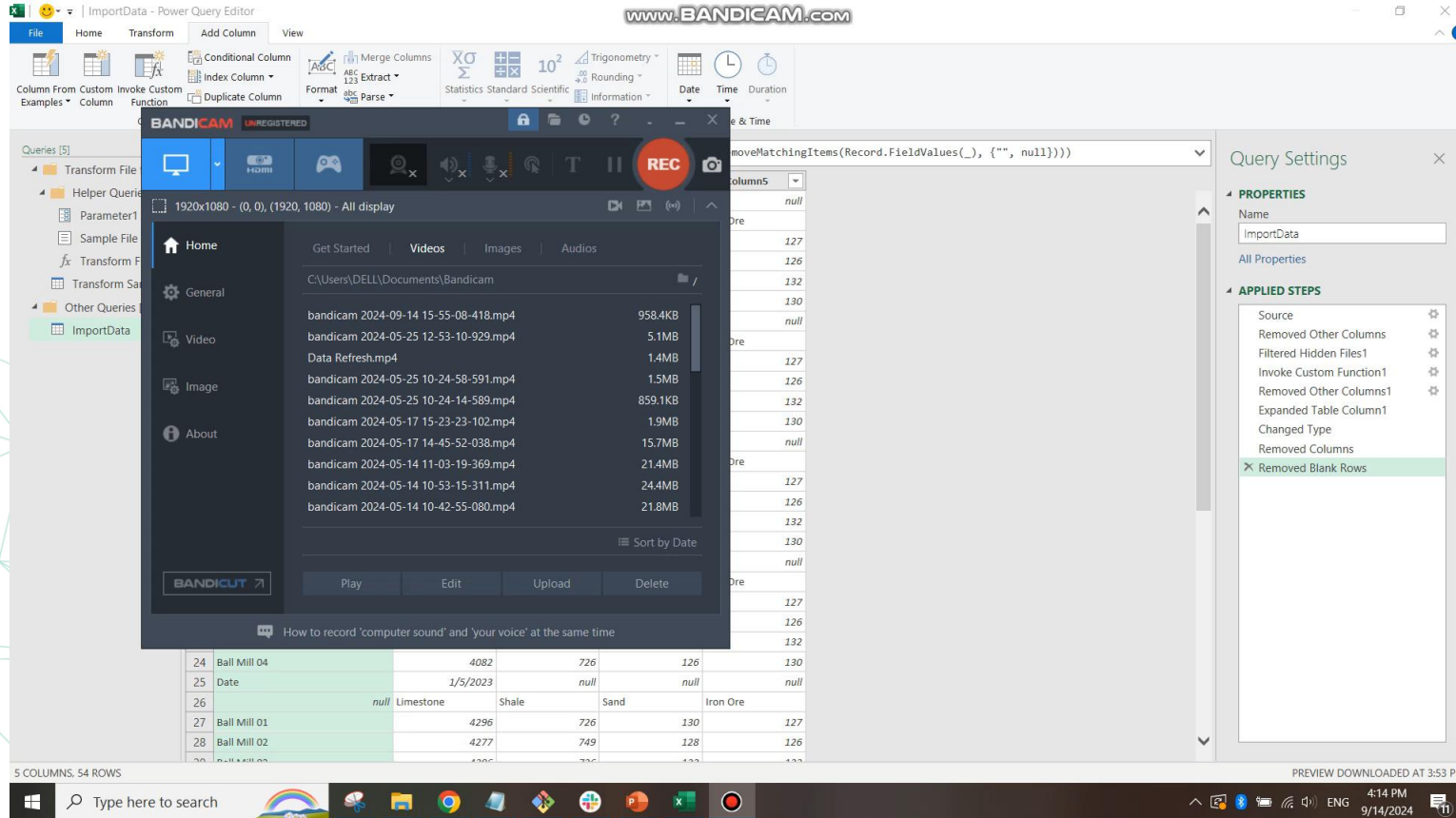
Follow from the Video



The screenshot displays the Power Query Editor interface. The main area shows a data table with columns and rows. A 'Query Settings' pane on the right indicates the 'APPLIED STEPS' list, which includes 'Changed Type'. A 'BANDICAM' watermark is visible in the top right corner of the editor.

Column1	Column2	Column3	Column4	Column5	Column6
24	Ball Mill 04	4082	726	126	127
25					126
26	Date	1/4/2023			132
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Follow from the Video



There are two main ways to load queries into your workbook:

- From the **Power Query Editor**, you can use the **Close and Load** commands in the **Close** group on the **Home** tab.
- From the **Excel Workbook Queries** pane (Select **Queries & Connections**), you can right-click a query and select **Load To**.

You can also fine-tune your load options by using the **Query Options** dialog box (Select **File > Options and Settings > Query Options**) to choose how you want to view your data and where you want to load the data, either in a worksheet or a **Data Model** (which is a relational data source of multiple tables that reside in a workbook).

Create, load, or edit a query in Excel (Power Query)

Create a Query

- You can either create a query from imported data or start with a blank query.

- Create a Query from Imported Data**

- This is the most common way to create a query.

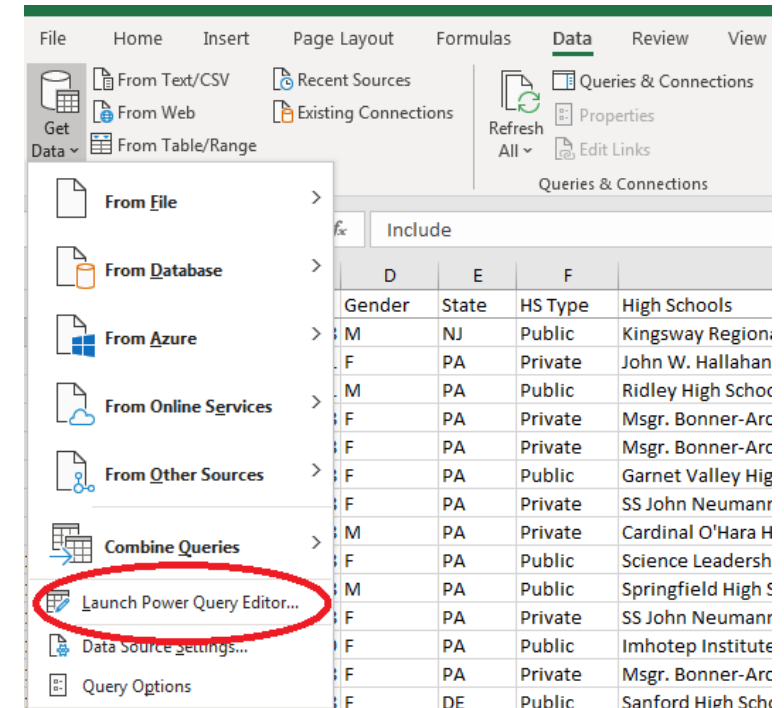
- Import some data.
- Select a cell in the data and then select **Query > Edit**.

- Create a Blank Query**

You might want to start from scratch. There are two ways to do this:

- Select **Data > Get Data > From Other Sources > Blank Query**.
- Select **Data > Get Data > Launch Power Query Editor**.

- Select New Source** to add a data source. This command is similar to the **Data > Get Data** command in the Excel ribbon.
- Select Recent Sources** to choose from a data source you have been working with. This command mirrors the **Data > Recent Sources** command in the Excel ribbon.
- Select Enter Data** to manually enter data. You might choose this command to experiment with the Power Query Editor independently of an external data source.



Load a Query

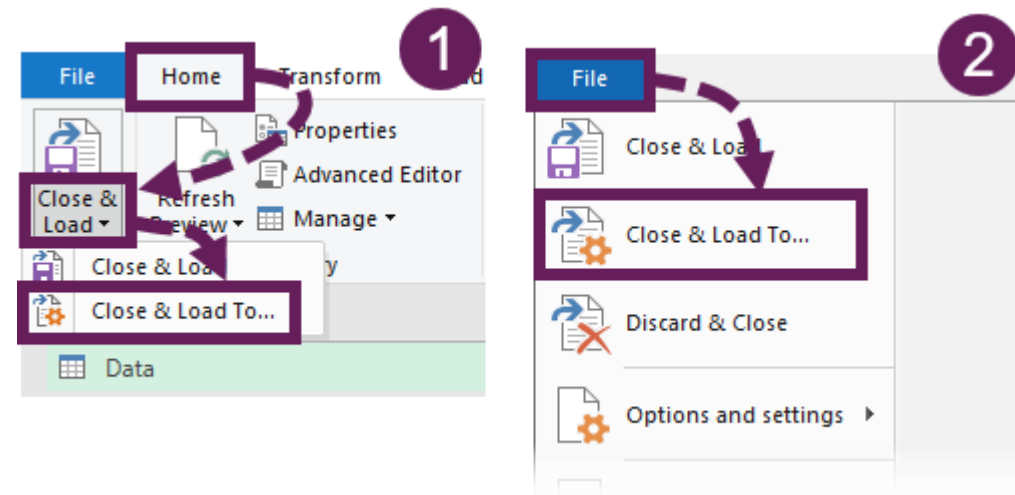
Assuming your query is valid and has no errors, you can load it back to a worksheet or Data Model.

Load a Query from the Power Query Editor

In the Power Query Editor, do one of the following:

- To load to a worksheet, select **Home > Close & Load > Close & Load**.
- To load to a Data Model, select **Home > Close & Load > Close & Load To**. In the **Import Data** dialog box, select **Add this data to the Data Model**.

In the Import Data dialog box, select Add this data to the Data Model.

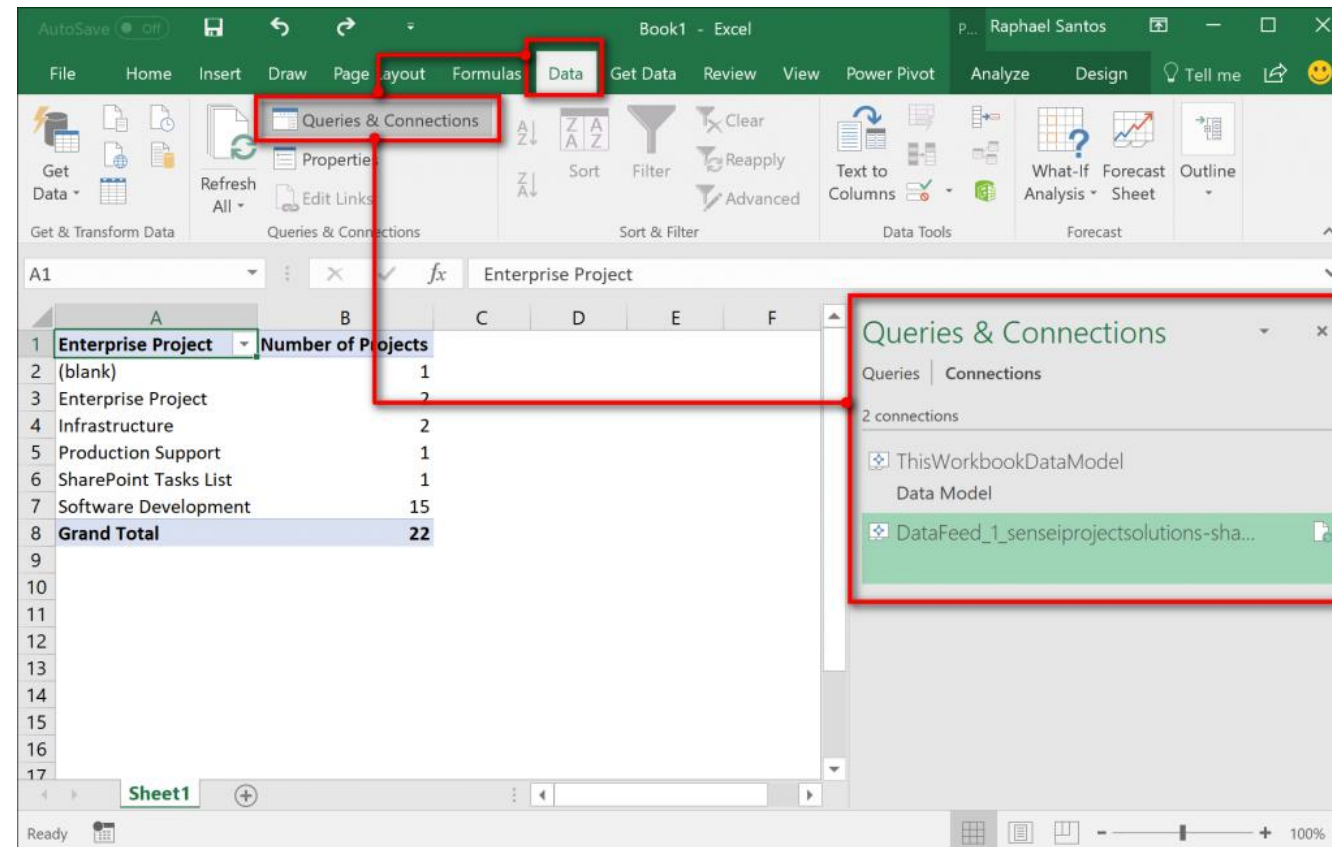


Load a Query

Load a Query from the Queries and Connections Pane

In Excel, you may want to load a query into another worksheet or Data Model. Follow these steps:

1. Select **Data > Queries & Connections**, and then select the **Queries** tab.
2. In the list of queries, locate the query, right-click on it, and then select **Load To**. The **Import Data** dialog box appears.
3. Decide how you want to import the data, and then select **OK**. For more information about using this dialog box, select the question mark (?).



Edit a Query from a Worksheet

To edit a query, locate one previously loaded from the Power Query Editor, select a cell in the data, and then select **Query > Edit**.

Edit a Query from the Queries & Connections Pane

You may find the **Queries & Connections** pane more convenient when you have many queries in one workbook and want to quickly find one.

1. In Excel, select **Data > Queries & Connections**, and then select the **Queries** tab.
2. In the list of queries, locate the query, right-click on it, and then select **Edit**.

Edit a Query from the Query Properties Dialog Box

1. In Excel, select **Data > Data & Connections > Queries** tab, right-click on the query, and select **Properties**.
2. In the **Properties** dialog box, select the **Definition** tab, and then select **Edit Query**.

Common Errors to Avoid

- **Mismatched Data Types:**
 - **Tip:** Verify and set correct data types for columns before transformation to avoid errors.
- **Duplicate Column Names:**
 - **Tip:** Rename columns to prevent confusion and maintain clarity.
- **Broken Queries:**
 - **Tip:** Refresh the query after edits to ensure all steps work correctly.
- **Incorrect Step Order:**
 - **Tip:** Review and rearrange steps in the Applied Steps pane for logical flow.

Q&A

Task

- **Task:**

- Import **sales data** from multiple CSV files using **Power Query**.
- **Clean the data** by removing duplicates, fixing data types, and filling in missing values.
- Merge the CSV files into a **single dataset**.
- Transform the dataset to include **new calculated columns** for total sales and profit margin.

- **Challenge:**

- Set up the Power Query steps so that you can **automate the entire cleaning and transformation process** for future data imports.

Thank You